Florian E. C. Blanc, PhD

CNRS Research Scientist in computational molecular biophysics

florian.blanc@cnrs.fr | French | November 19, 1990 | https://flblanc.github.io

Research interests

Protein dynamics, molecular machines & motors, integrative MD simulations, free energy calculations, machine learning, statistical mechanics

Current position

CNRS Research Scientist — *Villeurbanne (Lyon), France*

Biophysics of Complex Systems team - Institut des Sciences Analytiques — Since October 2024

Permanent independent position.

Research experience

Post-doctoral researcher — *Villeurbanne (Lyon), France*

Biophysics of Complex Systems team - Institut des Sciences Analytiques — September 2023 - September 2024

Integrative molecular modelling of bacterial silver-resistance proteins
Advisors: Olivier Walker, Maggy Hologne

Post-doctoral researcher — *Frankfurt am Main, Germany*

Department of Theoretical Biophysics - Max-Planck Institute of Biophysics — May 2019 - August 2023

Computational study of ATPase molecular machines
Advisor: Gerhard Hummer

Post-doctoral researcher — *Strasbourg, France*

Molecular Function and Design team - Université de Strasbourg — October 2018 - March 2019

Free energy simulations of biological and synthetic molecular machines
Advisor: Marco Cecchini

PhD candidate — *Strasbourg and Paris, France*

Molecular Function and Design team - Université de Strasbourg; Structural Motility team - Institut Curie — September 2014 - September 2018

Exploring chemo-mechanical transduction in the myosin molecular motor through computer simulations
Advisors: Marco Cecchini, Anne Houdusse

Funding and resources

DARI/GENCI allocation

PI — May 2026 - April 2027

ANR MyoAssemble

Partner — January 2026 - December 2029

Supervision

Junghwan Ro — M1 student, Chimie et Science des Matériaux, Université Claude Bernard Lyon 1, May - July 2026: Lignin characterization by NMR and MD simulations (Co-supervisor, with Hugo Lilti, Léa Vilcocq (IRCELYON))
Raphael Desrues — PhD student, Institut des Sciences Analytiques (ISA) / Université Claude Bernard Lyon 1, since October 2026: Molecular mechanisms of Myosin 18 regulatory functions from molecular simulations (Main supervisor, with Olivier Walker); Research internship (after an MSc in In Silico Drug Design, Université Paris Cité), February - July 2026: Gas-phase oligomerization of peptides (Co-supervisor, with Fabien Chirot (ILM))
Rim Annad — M1 student, Bio-informatique, Université Clermont Auvergne, May - July 2025: Coarse-grained simulations of an intrinsically disordered metalloprotein (Main supervisor, with Olivier Walker)
Matthieu Andriamisetra — L3 student, Sciences de la Vie, parcours Biochimie, Université Claude Bernard Lyon 1, March - May 2025: Kinetic analysis metal-induced peptide folding (Main supervisor, with Olivier Walker)
Mélodie Demontrond — M1 student, Analyse et Contrôle, Université Claude Bernard Lyon 1, March - May 2024: Machine learning analysis of peptide simulations (Main supervisor, with Olivier Walker)
Fanny Cuesta — Master's thesis (part-time), Biophysics, Goethe University Frankfurt, February 2022 - April 2025: Computational study of ATP Synthase c-ring and central stalk (Main supervisor, with Gerhard Hummer)

Student project supervision

Louison Grizard, Axel Lukaszewicz — Engineering students, CPE Lyon, April - May 2026 (19 days) (Supervisor)
Victor Boudet, Enzo Guedes — Engineering students, CPE Lyon, April - May 2025 (19 days) (Supervisor)

Collective responsibilities

AI advisor ("référent IA")

ISA — *Lyon, France* — Current

ISA representative for the Digital arm (Axe Numérique) of the IMPULSE Carnot institute

Lyon, France — Current

Elected Post-doc representative

Max Planck Institute of Biophysics — *Frankfurt-am-Main, Germany* — November 2020 - August 2023

Reviewing

Reviewer for PNAS, PCCP, Frontiers in Physiology, Comm. Chem., Nat. Comput. Sci., JCTC, BBRC
Scientific reviewer for the ECOS programme (2025)

Education

PhD in Theoretical and Computational Chemistry

Strasbourg and Paris, France
Institut de Sciences et d'Ingénierie Supramoléculaires / Université de Strasbourg / Institut Curie — September 2014 - September 2018
Highest rating ("exceptional") on all evaluation criteria.

MSc in Bioinformatics

Paris, France
Université Paris-Diderot — September 2013 - June 2014
With high honours.

École Normale Supérieure Diploma in Biology (major) and Physics (minor)

Paris, France
École Normale Supérieure (ENS) — September 2010 - June 2014

Admission to ENS - Biology Department

Nationwide competitive entrance evaluation - rank: 1. Admitted to ENS with the status of paid civil servant (élève-normalien). — July 2010

Classes Préparatoires BCPST

Marseille, France
Preparatory classes for nationwide competitive admission exams to French "Grandes Écoles" - 2-year intensive training in biology, chemistry, physics, Earth sciences and mathematics — September 2008 - June 2010

Baccalauréat Scientifique

Toulon, France

French High-School diploma, scientific specialization

June 2008

With highest honours.

Skills

Communication	Oral and written scientific communication
Languages	French (native), English (fluent), German (A1.2)
Programming	Python/SciPy, bash, Fortran 90 (basic knowledge), Tcl (basic knowledge)
Molecular Simulations	Molecular Dynamics, free energy calculations, enhanced sampling, quantum chemistry
Machine Learning	Generative deep learning, dimensionality reduction, clustering

Teaching duties

Qualification Universitaire (2020-2028) in Sections 31 (Theoretical, Physical, Analytical Chemistry) and 64 (Biochemistry, Molecular Biology)

Dynamique moléculaire et champs de forces (Molecular dynamics and force fields)

Guest lecture, Master 1 Chemistry, Université Claude Bernard Lyon 1

March 19, 2026

Intelligence artificielle et recherche en chimie (Artificial intelligence and research in chemistry)

Guest lecture, Master 1 Chemistry, Université Claude Bernard Lyon 1

December 1, 2025

Computational drug discovery

Frankfurt am Main, Germany

Graduate course, Goethe University Frankfurt

2020-2021 Winter semester

- Prepared and gave a 12 hour lecture series on **Advanced Simulation Methods for Drug Design**
- Invited a guest lecturer from the private sector
- Supervised and graded practicals and student presentations

Méthodologie Scientifique

Strasbourg, France

Undergraduate course, Université de Strasbourg

September 2017

- Taught a 1-week tutorial class on **Mathematical Methods for Chemistry** to first-year physical chemistry undergraduates

Biologie Structurale et Modélisation

Strasbourg, France

Graduate course, Université de Strasbourg

November 2016

- Gave a 2-hour **Statistical Mechanics** lecture in the Chemoinformatics Master program
- Supervised a 4-hour lab session on **Normal Mode Analysis** in the Chemoinformatics Master program

Traitement stochastique du signal biomédical

Noisy le Grand, France

Graduate course, Institut Supérieur des BioSciences (ISBS)

December 2014 & December 2015

- Supervised a 8-hour lab session on **Biomedical Signal Acquisition and Processing**

Volunteer mentor for high-school students

Paris, France

Mentoring, TalENS (ENS student association for high-school student mentoring)

September 2010 - June 2014

- Mentored and taught scientific concepts to high-school students
- Designed transdisciplinary scientific classes

Publications

☉ Trainee under my supervision

Metadynamics Simulations Meet Ligand Design for the Reversible Inhibition of Human Peroxiredoxin 5
Laura Troussicot, Florian E. C. Blanc, Yoann Pascal, Sebastien Vidal, Jean-Marc Lancelin, Florence Guillière
ACS Omega, 2026
https://doi.org/10.1021/acsomega.6c04953

Mechanism of Ag^+-Induced Folding of a Bacterial Peptide from Replica-Exchange Molecular Simulations
Florian E. C. Blanc (corr.), Maggy Hologne, Marion Girod, Mélodie Demontrond☉, Henry Chermette, Olivier Walker (corr.)
The Journal of Physical Chemistry Letters, 2026
https://doi.org/10.1021/acs.jpclett.6c00044

A Weak Coupling Mechanism for the Early Steps of the Recovery Stroke of Myosin VI: A Free Energy Simulation and String Method Analysis
Florian E. C. Blanc (corr.), Anne Houdusse, Marco Cecchini (corr.)
PLOS Computational Biology, 2024
https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1012005

Mechanism of Proton-Powered c-Ring Rotation in a Mitochondrial ATP Synthase
Florian E. C. Blanc, Gerhard Hummer
Proceedings of the National Academy of Sciences, 2024
https://doi.org/10.1073/pnas.2314199121

Molecular Mechanisms of Inorganic-Phosphate Release from the Core and Barbed End of Actin Filaments
Wout Oosterheert*, Florian E. C. Blanc*, Ankit Roy, Alexander Belyy, Micaela Boiero Sanders, Oliver Hofnagel, Gerhard Hummer, Peter Bieling, Stefan Raunser (* equal contribution)
Nature Structural & Molecular Biology, 2023
https://doi.org/10.1038/s41594-023-01101-9

Antibody Accessibility Determines Location of Spike Surface Mutations in SARS-CoV-2 Variants
Sören von Bülow, Mateusz Sikora, Florian E. C. Blanc, Roberto Covino, Gerhard Hummer
PLOS Computational Biology, 2023
https://doi.org/10.1371/journal.pcbi.1010822

Computational Epitope Map of SARS-CoV-2 Spike Protein
Mateusz Sikora*, Sören von Bülow*, Florian E. C. Blanc*, Michael Gecht*, Roberto Covino*, Gerhard Hummer (* equal contribution)
PLOS Computational Biology, 2021
https://doi.org/10.1371/journal.pcbi.1008790

An Asymmetric Mechanism in a Symmetric Molecular Machine
Florian E. C. Blanc, Marco Cecchini
The Journal of Physical Chemistry Letters, 2021
https://doi.org/10.1021/acs.jpclett.1c00404

In Situ Structural Analysis of SARS-CoV-2 Spike Reveals Flexibility Mediated by Three Hinges
Beata Turoňová*, Mateusz Sikora*, Christoph Schürmann*, Wim J. H. Hagen, Sonja Welsch, Florian E. C. Blanc, Sören von Bülow, Michael Gecht, Katrin Bagola, Cindy Hörner, Ger van Zandbergen, Jonathan Landry, Nayara Trevisan Doimo de Azevedo, Shyamal Mosalaganti, Andre Schwarz, Roberto Covino, Michael D. Mühlebach, Gerhard Hummer, Jacomine Krijnse Locker, Martin Beck (* equal contribution)
Science, 2020
https://doi.org/10.1126/science.abd5223

An Intermediate Along the Recovery Stroke of Myosin VI Revealed by X-Ray Crystallography and Molecular Dynamics
Florian E. C. Blanc, Tatiana Isabet, Hannah Benisty, H. Lee Sweeney, Marco Cecchini, Anne Houdusse
Proceedings of the National Academy of Sciences, 2018
https://doi.org/10.1073/pnas.1711512115

The Myosin X Motor Is Optimized for Movement on Actin Bundles
Virginie Ropars*, Zhaohui Yang*, Tatiana Isabet*, Florian Blanc, Kaifeng Zhou, Tianming Lin, Xiaoyan Liu, Pascale Hissier, Frédéric Samazan, Béatrice Amigues, Eric D. Yang, Hyokeun Park, Olena Pylypenko, Marco Cecchini, Charles V. Sindelar, H. Lee Sweeney, Anne Houdusse (* equal contribution)
Nature Communications, 2016
https://doi.org/10.1038/ncomms12456

Myosin MyTH4-FERM Structures Highlight Important Principles of Convergent Evolution
Vicente José Planelles-Herrero, Florian Blanc, Serena Sirigu, Helena Sirkia, Jeffrey Clause, Yannick Sourigues, Daniel O. Johnsrud, Beatrice Amigues, Marco Cecchini, Susan P. Gilbert, Anne Houdusse, Margaret A. Titus
Proceedings of the National Academy of Sciences, 2016
https://doi.org/10.1073/pnas.1600736113

Communications

Invited talks

Unveiling functional mechanisms of ATPase proteins with enhanced sampling simulations
Invited speaker, CECAM Workshop "Integrative and Multiscale Modelling Approaches to Illuminate CryoEM/ET data", Toulouse, March 2026

Unveiling Complex Biomolecular Processes with Enhanced Sampling Molecular Dynamics
Invited seminar, Centre de RMN à Très Hauts Champs (CRMN), Lyon, November 2025 (invited by Anne Lesage)

Revealing complex biomolecular processes with enhanced sampling Molecular Dynamics simulations
Invited seminar, Department of Chemistry, ENS Lyon, July 2025 (invited by Natacha Gillet)

Exploring the functional mechanisms of biomolecular machines with atomically-detailed simulations
Invited seminar, Laboratoire de Biologie et Modélisation de la Cellule (LBMC), ENS Lyon, November 2023 (invited by Riccardo Pellarin)

Exploring the function of molecular machines with simulations
Invited seminar, Institut des Sciences Analytiques, Villeurbanne, November 2021 (invited by Olivier Walker)

Oral communications

Mechanism of Ag^+-induced folding of a bacterial peptide from molecular dynamics simulations and deep dimensionality reduction
Machine Learning in Chemistry and Physics (MLCP2025) meeting, Lyon, France, June 2025

Mechanism and energetics of proton-powered c-ring rotation in mitochondrial ATP synthase
Congrès du Groupe de Graphisme et Modélisation Moléculaires (GGMM), Lille, France, September 2021

Poster presentations

Mechanism of Ag^+-induced folding of a bacterial peptide from molecular dynamics simulations
Blanc, F. E. C.; Hologne, M.; Demontrond, M.; Chermette, H.; Walker, O.
Congrès du GGMM, Forges-les-Eaux, France, 2025

Mechanism of Ag^+-induced folding of a bacterial peptide from molecular dynamics simulations
Blanc, F. E. C.; Hologne, M.; Demontrond, M.; Chermette, H.; Walker, O.
ML4NGP (Machine Learning for Non Globular Proteins) meeting, Vilnius, Lithuania, 2025

A novel intermediate along the recovery stroke of Myosin VI revealed by X-ray crystallography and molecular dynamics
Florian Blanc, Tatiana Isabet, Hannah Benisty, H. Lee Sweeney, Marco Cecchini, Anne Houdusse
Congrès du Groupe de Graphisme et Modélisation Moléculaires (GGMM), Reims, France, 2017

A novel intermediate along the recovery stroke of Myosin VI revealed by X-ray crystallography and molecular dynamics
Florian Blanc, Tatiana Isabet, Hannah Benisty, H. Lee Sweeney, Marco Cecchini, Anne Houdusse
EMBO Meeting, Mannheim, Germany, 2016